

Comprehensive 2027 Division C Science Olympiad Scope and Data Architecture Report

The 2027 Science Olympiad competition season for Division C (grades 9–12) comprises 23 national official events across five core STEM domains: Life, Personal & Social Science; Earth and Space Science; Physical Science & Chemistry; Technology & Engineering; and Inquiry & Nature of Science. To establish a comprehensive content architecture for SciOly.app, every event must be rigorously evaluated for knowledge-testing components. Knowledge-testing components include pure written exams, station-based practical exams, build-test hybrids where device performance is paired with a physics exam, and performance-knowledge hybrids.

Event Inclusion and Exclusion Determination

The determination of whether a Division C event is included in the knowledge-testing data structure depends on the presence of an academic or written evaluation. Events that consist strictly of mechanical construction, structural load testing, or flight duration trials without a knowledge-testing component are excluded from detailed study taxonomy profiling.

Event Name	Domain	Included?	Categorization / Exclusion Rationale
Anatomy and Physiology	Life, Personal & Social Science	Yes	Pure written / station practical exam.
Astronomy	Earth and Space Science	Yes	Pure written test on astrophysics and stellar evolution.
Boomilever	Technology & Engineering	No	Pure structural engineering build; scored strictly on load-supported structural efficiency.
Botany	Life, Personal & Social Science	Yes	Pure written / station exam (new official event replacing Entomology).
Chemistry Lab	Physical Science & Chemistry	Yes	Lab practical with written analysis and theoretical exam.
Circuit Lab	Physical Science & Chemistry	Yes	Lab practical with theoretical physics written exam.
Codebusters	Inquiry & Nature of Science	Yes	Pure written cryptanalysis exam / speed decryption.

Event Name	Domain	Included?	Categorization / Exclusion Rationale
Designer Genes	Life, Personal & Social Science	Yes	Pure written exam on molecular biology and genetics.
Disease Detectives	Life, Personal & Social Science	Yes	Pure written exam on epidemiology and biostatistics.
Dynamic Planet	Earth and Space Science	Yes	Pure written / station practical exam.
Electric Vehicle	Technology & Engineering	No	Pure mechanical build; scored on target distance accuracy, speed, and braking.
Engineering CAD	Inquiry & Nature of Science	Yes	Hybrid performance + computer drafting knowledge test.
Experimental Design	Inquiry & Nature of Science	Yes	Lab practical + written experimental design report analysis.
Forensics	Physical Science & Chemistry	Yes	Lab practical with qualitative chemical analysis exam.
Hovercraft	Physical Science & Chemistry	Yes	Build + written test on fluid mechanics and physics principles.
Mission Possible	Technology & Engineering	No	Pure mechanical build; scored on Rube Goldberg task execution and timing.
Ping-Pong Parachute	Inquiry & Nature of Science	No	Pure pressure-launch flight event; scored on flight duration.
Protein Modeling	Physical Science & Chemistry	Yes	Hybrid performance + physical model construction + written exam.
Remote Sensing	Earth and Space Science	Yes	Pure written / station test on satellite physics and image analysis.
Rocks and Minerals	Earth and Space Science	Yes	Station practical test on petrology, mineralogy, and economic geology.
Thermodynamics	Physical Science & Chemistry	Yes	Build + written test on thermodynamic laws and heat transfer.
Water Quality	Life, Personal & Social	Yes	Lab practical with

Event Name	Domain	Included?	Categorization / Exclusion Rationale
	Science		written analysis exam + salinometer test.
Wright Stuff	Technology & Engineering	No	Pure rubber-band flight build; scored on duration in flight.

PART 1 — MASTER TABLE

The master table below summarizes all 18 qualifying Division C events for the 2027 season, outlining test components, explicit official topics, subtopic decompositions, concept structures, simplified competition materials, primary source links, and confidence notes.

Event Name	Type of Test Component	Official Topic(s)	Subtopics	Concepts	Allowed Materials	Best Source Links	Confidence Notes
Anatomy and Physiology	pure written test / station practical	Cardiovascular, Lymphatic, Excretory systems	Anatomy of Heart & Vessels, Cardiac Physiology, Lymphatics & Immunity, Renal Anatomy, Nephron Physiology	Heart chambers, Cardiac cycle, Blood pressure regulation, Innate immunity, Glomerular filtration, Acid-base balance	Binder, Calculator	[soinc.org/anatomy-and-physiology-c](https://www.soinc.org/events/2027-division-c-events)	Directly official 2027 Year 4 rotation (Cardio/Lymph/Excretory).
Astronomy	pure written test	Stellar evolution: formation to destruction, HR diagram, active galactic nuclei	Stellar Formation & Main Sequence, Late-Stage Evolution, Binary Systems, Extragalactic Astrophysics, Astrophysics Math	Protostars, Jeans Mass, Supernovae, Degeneracy pressure, AGN, Redshift, Distance modulus, Kepler's laws	Binder, Calculator	[soinc.org/astronomy-c](https://www.soinc.org/events/2027-division-c-events)	Directly official core astrophysics scope.
Botany	pure written test / station practical	Plant anatomy, physiology, taxonomy, diseases, and	Anatomy & Morphology, Plant Physiology, Major Plant Groups,	Xylem/phloem, Photosynthesis (C3/C4/CAM),	Cheat sheet, Calculator	scioly.org/wiki/Botany	Directly official 2027 new event replacing Entomology

Event Name	Type of Test Component	Official Topic(s)	Subtopics	Concepts	Allowed Materials	Best Source Links	Confidence Notes
		horticulture	Pathology & Deficiencies, Horticulture	Bryophytes /Angiosperms, Pathogens, NPK deficiency			y.
Chemistry Lab	station/lab practical with written analysis	Gas Laws, Kinetics	Gas Laws & Real Gases, Kinetic Molecular Theory, Reaction Rates, Rate Laws, Reaction Mechanisms	Ideal gas law, van der Waals equation, Effusion, Rate law determination, Arrhenius equation, Catalysts	Cheat sheet, Calculator, Lab equipment	soinc.org/chem-lab-c	Directly official 2027 topic rotation (Gas Laws, Kinetics).
Circuit Lab	station/lab practical with written analysis	DC & AC circuits, conductors, semiconductors, digital logic	DC Circuit Analysis, AC Circuit Principles, Reactive Components, Semiconductors, Digital Logic	Kirchhoff's laws, Thevenin equivalents, RLC impedance, Diodes/Tra nsistors, Logic gates, Truth tables	Binder, Calculator, Multimeter, Lab tools	soinc.org/circuit-lab-c	Directly official physical science core scope.
Codebusters	pure written test	Cryptanalysis of historic ciphers, Hill ciphers, Affine ciphers	Monoalphabetic Substitutions, Polyalphabetic Ciphers, Mathematical Ciphers, Transposition Ciphers, Speed Analysis	Aristocrats, Patristocrats, Xenocrypts, Vigenère, Hill Ciphers, Affine Ciphers, Baconian, Rail Fence	Calculator, Pencils	scioly.org/wiki/Codebusters	Directly official 2026–2027 cipher rules additions.
Designer Genes	pure written test	Classical genetics,	Mendelian Inheritance,	Pedigree analysis,	Binder, Calculator	soinc.org/designer-genes	Directly official

Event Name	Type of Test Component	Official Topic(s)	Subtopics	Concepts	Allowed Materials	Best Source Links	Confidence Notes
		molecular biology, biotechnology, genomics	Molecular Genetics, Gene Regulation, Recombinant DNA, Population Genetics	Transcription/Translation, Operons, PCR, CRISPR, Hardy-Weinberg equilibrium		nes-c	molecular genetics scope.
Disease Detectives	pure written test	Epidemiology, disease transmission, outbreak investigation, biostatistics	Outbreak Investigation Steps, Disease Transmission, Measures of Association, Diagnostic Testing, Interventions	10 steps of investigation, Attack rate, Odds ratio, Relative risk, Sensitivity/Specificity, Herd immunity	Cheat sheet, Calculator	soinc.org/division-c/disease-detectives-c	Directly official epidemiology framework.
Dynamic Planet	pure written test / station practical	Oceanography	Chemical & Physical Oceanography, Ocean-Atmosphere Coupling, Waves & Tides, Marine Geology, Climate Phenomena	Salinity/density, Ekman transport, Thermohaline circulation, Wave physics, Tides, Bathymetry, ENSO	Binder, Calculator	soinc.org/division-c/dynamic-planet-c	Directly official 2027 Oceanography rotation.
Engineering CAD	hybrid performance + knowledge test	3D CAD modeling, drafting standards, orthographic views, GD&T	Technical Drafting Standards, Parametric 3D Modeling, Assembly Design, GD&T,	Orthographic projections, Extrude/Revolve/Loft, Constraints, Mating, Tolerancing, Mass	Cheat sheet, Laptop, CAD software, Drafting tools	soinc.org/events/2027-division-c-engineering-cad-ventures	Directly official 2027 new national event.

Event Name	Type of Test Component	Official Topic(s)	Subtopics	Concepts	Allowed Materials	Best Source Links	Confidence Notes
			Physical Properties	calculations			
Experimental Design	station/lab practical with written analysis	Scientific method, variable isolation, data analysis, report writing	Problem & Hypothesis, Variable Isolation, Data Collection, Graphical Analysis, Statistics & Error Analysis	Independent/Dependent variables, Constants, Error analysis, Linear regression, Standard deviation	Binder, Calculator	soinc.org/experimental-design-c	Directly official experimental design rubric.
Forensics	station/lab practical with written analysis	Qualitative chemical analysis, polymers, physical evidence	Qualitative Analysis, Polymers & Fibers, Chromatography, Serology & DNA, Physical Evidence	Flame tests, Solubility, Plastic density, Ink chromatography, Electrophoresis, Fingerprints	Cheat sheet, Calculator, Lab equipment	soinc.org/forensics-c	Directly official forensics chemistry standards.
Hovercraft	build + written test	Fluid mechanics, pressure dynamics, impulse, friction, circuits	Fluid Statics, Aerodynamic Forces, Mechanics & Impulse, Device Electromechanics, Performance Calculations	Bernoulli's principle, Drag coefficient, Air cushion pressure, Thrust vs Lift motor power, Stopping impulse	Cheat sheet, Calculator, Built device	soinc.org/hovercraft-c	Directly official hybrid build/test event.
Protein Modeling	hybrid performance + knowledge test	CRISPR-Cas9 genome editing, base editors, protein	Amino Acid Chemistry, Protein Folding Levels, Visualization Software,	PAM sequences, Guide RNA design, Alpha helices/Beta sheets,	Binder, Pre-built model, Computer/Software	soinc.org/protein-modeling-c	Directly official 2027 CRISPR-Cas9 topic.

Event Name	Type of Test Component	Official Topic(s)	Subtopics	Concepts	Allowed Materials	Best Source Links	Confidence Notes
		folding	CRISPR-Cas9, Biochemistry, Base Editing	PyMOL commands, Cytidine deaminases			
Remote Sensing	pure written test / station practical	Satellite imagery, EM spectrum, radiation laws, climate observation	Radiation Physics, Satellite Sensors & Orbits, Image Processing, Spectral Indices, Climate Applications	Planck's Law, Wien's Law, Sun-synchronous orbits, NDVI calculation, Band ratios, Hyperspectral imaging	Binder, Calculator	soinc.org/remote-sensing-c	Directly official remote sensing physics.
Rocks and Minerals	station/lab practical with written analysis	Mineralogy, crystallography, petrology, economic geology	Crystallography & Minerals, Igneous Petrology, Sedimentary Petrology, Metamorphic Petrology, Economic Geology	Crystal systems, Mohs scale, Bowen's Reaction Series, Foliation, Ore deposits, Stratigraphy	Binder, Calculator, Magnifying glass	soinc.org/rocks-and-minerals-c	Directly official rock and mineral list.
Thermodynamics	build + written test	Laws of thermodynamics, heat transfer, cycles, calorimetry	Thermal Properties, Heat Transfer Mechanisms, Thermodynamic Laws, Processes & Cycles, Calorimetry	Conduction, Convection, Radiation, Enthalpy, Entropy, Carnot efficiency, Heat capacity, R-values	Cheat sheet, Calculator, Built device	soinc.org/thermodynamics-c	Directly official hybrid thermodynamics event.
Water Quality	station/lab practical with written	Marine and Estuary ecosystem	Marine & Estuarine Ecology,	Estuary zones, Dissolved	Cheat sheet, Calculator,	scioly.org/wiki/Water_Quality	Directly official 2027

Event Name	Type of Test Component	Official Topic(s)	Subtopics	Concepts	Allowed Materials	Best Source Links	Confidence Notes
	analysis	s, water chemistry, salinometer	Water Chemistry, Bioindicators, Hydrology & Pollution, Salinometer Physics	oxygen, BOD, Eutrophication, Macroinvertebrates, Archimedes' Principle	Built salinometer		Marine/Estuary rotation.

DETAILED EVENT PROFILES

Anatomy and Physiology

B. Official 2027 scope

- Exact 2027 scope summary: Focuses on human anatomy and physiology, rotating to the Cardiovascular, Lymphatic, and Excretory systems for the 2027 competition season.
- Official topics explicitly named: Cardiovascular System, Lymphatic System and Immunity, Excretory System, associated cellular mechanisms, and major clinical disorders.

C. Subtopic map

- Cardiovascular System Anatomy and Histology
- Cardiac Physiology and Hemodynamics
- Lymphatic System and Innate Immunity
- Adaptive Immunity and Immunoglobulins
- Excretory System Anatomy and Renal Physiology

D. Concept graph

- Cardiovascular System Anatomy and Histology: Gross anatomy of heart chambers, cardiac valves, coronary circulation, histology of arterial, venous, and capillary walls, hematology components (erythrocytes, leukocytes, thrombocytes), clotting cascade.
- Cardiac Physiology and Hemodynamics: Cardiac conduction system (SA node, AV node, Bundle of His, Purkinje fibers), electrocardiogram (ECG) wave components, cardiac cycle phases, Frank-Starling law, Poiseuille's law, blood pressure regulation via baroreceptors and chemoreceptors.
- Lymphatic System and Innate Immunity: Primary and secondary lymphoid organs (thymus, bone marrow, spleen, lymph nodes), lymphatic vessel network, physical and chemical innate barriers, inflammatory response, complement system activation pathways.
- Adaptive Immunity and Immunoglobulins: B cell activation and humoral immunity, T cell cell-mediated immunity (CD4+ helper vs CD8+ cytotoxic T cells), antigen presentation via MHC I and MHC II, antibody structures and immunoglobulin classes (IgA, IgD, IgE, IgG,

IgM).

- Excretory System Anatomy and Renal Physiology: Macro and microanatomy of kidneys and nephrons, glomerular filtration rate (GFR) autoregulation, tubular reabsorption and secretion dynamics, countercurrent multiplier mechanism, RAAS pathway, ADH fluid regulation, acid-base balance maintenance.

Astronomy

B. Official 2027 scope

- Exact 2027 scope summary: Covers stellar evolution from collapse to remnant formation, observational astrophysics, active galactic nuclei, and quantitative cosmological calculation.
- Official topics explicitly named: Stellar evolution, Hertzsprung-Russell diagram, star clusters, supernovae, compact objects, active galactic nuclei, extragalactic astronomy, and mathematical astrophysics.

C. Subtopic map

1. Stellar Formation and Main Sequence Evolution
2. Late-Stage Stellar Evolution and Remnants
3. Variable Stars and Binary Systems
4. Extragalactic Astrophysics and Cosmology
5. Observational Astronomy and Mathematical Physics

D. Concept graph

- Stellar Formation and Main Sequence Evolution: Jeans instability criteria, protostellar collapse, Hayashi tracks, proton-proton chain vs CNO cycle nuclear fusion, hydrostatic equilibrium, HR diagram main sequence turnoff analysis.
- Late-Stage Stellar Evolution and Remnants: Red giant branch dynamics, helium flash, planetary nebulae, electron degeneracy pressure, Chandrasekhar limit, Type Ia vs Type II supernovae, neutron star structure, Tolman-Oppenheimer-Volkoff limit, black hole event horizons and Schwarzschild radius.
- Variable Stars and Binary Systems: Cepheid variable period-luminosity relationships, RR Lyrae stars, eclipsing binary light curve interpretation, Roche lobe overflow, accretion disk dynamics.
- Extragalactic Astrophysics and Cosmology: Active Galactic Nuclei classifications (quasars, blazars, Seyfert galaxies), Hubble-Lemaître law, dark matter evidence via galaxy rotation curves, cosmic microwave background radiation features.
- Observational Astronomy and Mathematical Physics: Stefan-Boltzmann law, Wien's displacement law, distance modulus calculations, Kepler's third law calculations, spectroscopic redshift equations.

Botany

B. Official 2027 scope

- Exact 2027 scope summary: Evaluates plant biology, morphology, physiological pathways, major plant taxonomy, pathology, and home horticulture.
- Official topics explicitly named: Plant anatomy and histology, photosynthesis and transpiration, plant taxonomy, plant diseases, nutrient deficiencies, and propagation techniques.

C. Subtopic map

1. Plant Anatomy and Histology
2. Plant Physiology and Metabolism
3. Plant Taxonomy and Evolutionary Groups
4. Plant Pathology and Nutrient Deficiencies
5. Horticulture and Plant Propagation

D. Concept graph

- Plant Anatomy and Histology: Xylem (vessels, tracheids) and phloem (sieve tubes, companion cells) structure, apical and lateral meristems, leaf mesophyll architecture, stomatal complex dynamics, root histology (endodermis, Casparian strip, stele).
- Plant Physiology and Metabolism: Photosynthetic light reactions, Calvin cycle (C₃, C₄ spatial separation pathway, CAM temporal separation pathway, cohesion-tension transpiration theory, plant hormones (auxins, cytokinins, gibberellins, ABA, ethylene), photoperiodism.
- Plant Taxonomy and Evolutionary Groups: Non-vascular plants (Bryophytes), seedless vascular plants (Pteridophytes), Gymnosperms, Angiosperms (Monocots vs Eudicots differentiation), alternation of generations life cycles.
- Plant Pathology and Nutrient Deficiencies: Essential macronutrients (NPK, Ca, Mg, S) and micronutrients, deficiency chlorosis and necrosis patterns, fungal pathogens (powdery mildew, rusts, blights), bacterial pathogens (crown gall), viral pathogens (TMV).
- Horticulture and Plant Propagation: Sexual reproduction vs asexual propagation (cuttings, grafting, tissue culture), soil texture analysis, soil cation exchange capacity, greenhouse management.

Chemistry Lab

B. Official 2027 scope

- Exact 2027 scope summary: Combines theoretical problem sets and hands-on laboratory experiments rotating specifically to Gas Laws and Chemical Kinetics for the 2027 season.
- Official topics explicitly named: Ideal and Real Gas Laws, Kinetic Molecular Theory, Reaction Rates, Differential and Integrated Rate Laws, Arrhenius Equation, Catalysis.

C. Subtopic Map

1. Ideal and Real Gas Thermodynamics

2. Kinetic Molecular Theory and Gas Transport
3. Chemical Kinetics and Rate Law Determination
4. Temperature Dependence and Reaction Mechanisms
5. Laboratory Titrations and Gas Measurements

D. Concept Graph

- Ideal and Real Gas Thermodynamics: Ideal gas equation ($PV = nRT$), gas stoichiometry, Dalton's law of partial pressures, van der Waals equation for real gases, compressibility factor (Z).
- Kinetic Molecular Theory and Gas Transport: KMT postulates, root-mean-square velocity (v_{rms}), Maxwell-Boltzmann distribution curves, Graham's law of effusion and diffusion.
- Chemical Kinetics and Rate Law Determination: Differential vs integrated rate laws (zero, first, second order), reaction half-life mathematical formulations, initial rates method, spectrophotometry and Beer-Lambert law ($A = \epsilon b C$).
- Temperature Dependence and Reaction Mechanisms: Arrhenius equation ($k = A e^{-E_a/RT}$), activation energy calculation, transition state theory, elementary steps, rate-determining step identification, homogeneous vs heterogeneous catalysts.
- Laboratory Titrations and Gas Measurements: Eudiometer gas collection over water, vapor pressure corrections, iodine clock reaction techniques, spectrophotometer cuvette calibration.

Circuit Lab

B. Official 2027 scope

- Exact 2027 scope summary: Focuses on theoretical circuit analysis and practical laboratory measurement of direct current (DC) and alternating current (AC) electrical systems, active components, and digital logic.
- Official topics explicitly named: DC circuits, AC circuits, passive components, semiconductors, operational amplifiers, and digital logic gates.

C. Subtopic Map

1. Direct Current (DC) Circuit Analysis
2. Alternating Current (AC) and Reactive Components
3. Semiconductor Devices and Operational Amplifiers
4. Digital Electronics and Logic Systems
5. Laboratory Instrumentation and Measurement

D. Concept Graph

- Direct Current (DC) Circuit Analysis: Ohm's law, Kirchhoff's voltage and current laws (KVL/KCL), series and parallel resistor networks, Delta-Wye transformations, Thevenin and Norton equivalent circuits, maximum power transfer theorem, nodal and mesh analysis.
- Alternating Current (AC) and Reactive Components: Capacitance and inductance

equations, RC and RL time constants (τ), sinusoidal AC waveforms (RMS values), capacitive and inductive reactance (X_C , X_L), RLC series/parallel resonance (ω_0), complex impedance.

- Semiconductor Devices and Operational Amplifiers: P-N junction diode characteristics, Zener voltage regulation, BJT and MOSFET transistor operating regions, ideal operational amplifier configurations (inverting, non-inverting, summing, buffers).
- Digital Electronics and Logic Systems: Boolean algebra simplifications, De Morgan's laws, standard logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR), truth table synthesis, combinational logic (multiplexers, decoders), flip-flops.
- Laboratory Instrumentation and Measurement: Digital multimeter (DMM) operation for voltage, current, and resistance, oscilloscope timebase and triggering, Wheatstone bridge measurements, breadboard prototyping techniques.

Codebusters

B. Official 2027 scope

- Exact 2027 scope summary: Evaluates speed and accuracy in decrypting and encrypting historic and mathematical ciphers, featuring Hill and Affine ciphers for 2026–2027.
- Official topics explicitly named: Monoalphabetic substitution ciphers (Aristocrats, Patristocrats, Xenocrypts), polyalphabetic ciphers, mathematical ciphers (Affine, Hill), transposition ciphers, Baconian ciphers.

C. Subtopic Map

1. Monoalphabetic Substitution Ciphers
2. Polyalphabetic and Transposition Ciphers
3. Mathematical and Algebraic Ciphers
4. Specialized Cryptographic Codes
5. Timed Cryptanalysis Strategies

D. Concept Graph

- Monoalphabetic Substitution Ciphers: Aristocrat decryption using English letter frequency and pattern recognition, Patristocrat deciphering without spaces, Xenocrypt cryptanalysis using Spanish language conventions, K1/K2/K3 keyword reconstruction.
- Polyalphabetic and Transposition Ciphers: Vigenère cipher encryption/decryption using tabula recta, Rail Fence cipher deciphering with varied rail counts and offsets, Columnar transposition.
- Mathematical and Algebraic Ciphers: Affine cipher math ($E(x) = (ax + b) \pmod{26}$) and modular multiplicative inverse computation, Hill cipher 2×2 and 3×3 matrix multiplication and modular matrix inversion.
- Specialized Cryptographic Codes: Baconian cipher variant decoding, Pollux and Morbit cipher mapping from digits to Morse code, RSA encryption modular arithmetic basics.
- Timed Cryptanalysis Strategies: First-question timing bonus optimization, standard error threshold rules, digraph/trigraph frequency table application.

Designer Genes

B. Official 2027 scope

- Exact 2027 scope summary: Covers comprehensive molecular biology, classical Mendelian genetics, recombinant DNA biotechnology, gene regulation, and population genetics.
- Official topics explicitly named: Inheritance patterns, pedigree analysis, DNA replication, transcription, translation, gene regulation operons, biotechnology tools, and genomics.

C. Subtopic Map

1. Classical Transmission Genetics
2. Molecular Biology and Central Dogma
3. Eukaryotic and Prokaryotic Gene Regulation
4. Recombinant DNA and Biotechnology
5. Population Genetics and Genomics

D. Concept Graph

- Classical Transmission Genetics: Monohybrid and dihybrid crosses, incomplete dominance, codominance, sex-linked/influenced/limited traits, pedigree pattern deduction, genetic mapping via recombination frequencies.
- Molecular Biology and Central Dogma: DNA double helix structural antiparallelism, semi-conservative replication machinery (helicase, primase, DNA polymerase, ligase), transcription mechanics (promoters, pre-mRNA splicing, 5' cap, poly-A tail), translation machinery (ribosomes, tRNA charging, codon-anticodon pairing).
- Eukaryotic and Prokaryotic Gene Regulation: Prokaryotic operon models (lac inducible vs trp repressible), eukaryotic transcription factors and enhancers, epigenetic chromatin modification (histone acetylation vs DNA methylation), RNA interference (miRNA/siRNA).
- Recombinant DNA and Biotechnology: Restriction enzyme digests, plasmid cloning vectors, PCR thermal cycling, gel electrophoresis, Sanger sequencing, Next-Gen sequencing, CRISPR-Cas9 target editing, blotting techniques (Southern, Northern, Western).
- Population Genetics and Genomics: Hardy-Weinberg equilibrium calculations ($p^2 + 2pq + q^2 = 1$), evolutionary mechanics (genetic drift, bottleneck, founder effect, natural selection), BLAST search bioinformatic analysis.

Disease Detectives

B. Official 2027 scope

- Exact 2027 scope summary: Focuses on public health, epidemiological investigation methodology, disease transmission patterns, and biostatistical analysis.
- Official topics explicitly named: Outbreak investigation steps, modes of transmission, measures of association, diagnostic test evaluations, and public health intervention

strategies.

C. Subtopic Map

1. Epidemiological Principles and Outbreak Investigation
2. Disease Transmission Patterns and Dynamics
3. Biostatistics and Measures of Association
4. Diagnostic Testing Metrics and Public Health Surveillance
5. Prevention, Control, and Public Health Intervention

D. Concept Graph

- Epidemiological Principles and Outbreak Investigation: 10 steps of an outbreak investigation, epidemic curve creation and classification (point source, continuous source, propagated), case definitions (suspected, probable, confirmed), epidemiological triad (agent, host, environment).
- Disease Transmission Patterns and Dynamics: Direct vs indirect contact, droplet, airborne, vehicle, and vector-borne transmission, reservoirs, carriers, endemic vs epidemic vs pandemic states, incubation period calculation.
- Biostatistics and Measures of Association: Incidence vs prevalence rates, attack rates, relative risk (RR) calculations in cohort studies, odds ratio (OR) calculations in case-control studies, 2 times 2 contingency tables, attributable risk.
- Diagnostic Testing Metrics and Public Health Surveillance: Sensitivity ($TP / [TP + FN]$), specificity ($TN / [TN + FP]$), positive predictive value (PPV), negative predictive value (NPV), active vs passive vs sentinel surveillance systems.
- Prevention, Control, and Public Health Intervention: Primary, secondary, and tertiary prevention levels, herd immunity thresholds ($1 - 1/R_0$), isolation vs quarantine protocols, environmental control interventions.

Dynamic Planet

B. Official 2027 scope

- Exact 2027 scope summary: Covers Earth's hydrosphere processes, rotating specifically to Oceanography for the 2027 competition season.
- Official topics explicitly named: Chemical and physical oceanography, atmospheric-ocean coupling, wave mechanics, tides, marine geology, bathymetry, and climate oscillations.

C. Subtopic Map

1. Physical and Chemical Properties of Ocean Water
2. Ocean-Atmosphere Dynamics and Atmospheric Circulation
3. Wave Mechanics, Tides, and Coastal Morphology
4. Marine Geology and Basin Bathymetry
5. Climate Oscillations and Marine Environmental Issues

D. Concept Graph

- Physical and Chemical Properties of Ocean Water: Salinity units (ppt), density drivers (temperature, salinity, pressure), thermocline, halocline, and pycnocline profiles, dissolved gas dynamics, Carbonate Compensation Depth (CCD).
- Ocean-Atmosphere Dynamics and Atmospheric Circulation: Coriolis force deflection, Ekman spiral and transport, ocean gyre systems, thermohaline global conveyor belt circulation, coastal and equatorial upwelling/downwelling.
- Wave Mechanics, Tides, and Coastal Morphology: Wave parameters (wavelength, period, speed $C = L/T$), deep vs shallow water waves, wave refraction and shoaling, equilibrium vs dynamic tide theories, spring and neap tides, longshore currents, coastal erosional/depositional features.
- Marine Geology and Basin Bathymetry: Passive vs active continental margins, abyssal plains, trenches, mid-ocean ridges, seamounts and guyots, marine sediment classifications (terrigenous, biogenous, hydrogenous), sonar and satellite altimetry.
- Climate Oscillations and Marine Environmental Issues: El Niño-Southern Oscillation (ENSO) phases, Pacific Decadal Oscillation (PDO), ocean acidification chemistry, hypoxia dead zones, tsunami wave physics.

Engineering CAD

B. Official 2027 scope

- Exact 2027 scope summary: Evaluates proficiency in 3D Computer-Aided Design (CAD), drafting standards, orthographic projections, parametric modeling, and GD&T principles.
- Official topics explicitly named: Technical drafting rules, 3D parametric feature modeling, assembly mating constraints, geometric dimensioning and tolerancing, and physical property analysis.

C. Subtopic Map

1. Technical Drafting Standards and Projection Theory
2. Parametric 3D Part Modeling
3. Assembly Design and Constraint Mating
4. Geometric Dimensioning and Tolerancing (GD&T)
5. Physical Properties Analysis and Model Output

D. Concept Graph

- Technical Drafting Standards and Projection Theory: First-angle vs third-angle orthographic projections, standard multi-view alignment (front, top, side), isometric views, full/half/offset sectional views, auxiliary views.
- Parametric 3D Part Modeling: 2D sketch geometric constraints (concentric, parallel, tangent, perpendicular), 3D operations (extrude, revolve, sweep, loft), detail features (fillet, chamfer, shell, pattern), parametric feature tree hierarchy.
- Assembly Design and Constraint Mating: Bottom-up vs top-down assemblies, assembly

mates (coincident, concentric, distance, parallel, angle), degrees of freedom (DOF) analysis, clearance and interference detection.

- Geometric Dimensioning and Tolerancing (GD&T): ASME Y14.5 standards, feature control frames, datum reference frames, tolerance categories (form, orientation, location, runout), MMC/LMC conditions.
- Physical Properties Analysis and Model Output: Material density assignment, volume, mass, surface area, and center of mass calculations, file export formats (.STEP, .STL, .DXF), slicer preparation.

Experimental Design

B. Official 2027 scope

- Exact 2027 scope summary: Assesses student ability to design, conduct, and report an empirical experiment on-site using provided raw materials.
- Official topics explicitly named: Problem statement, hypothesis creation, variable controls, experimental setup, data collection, graphing standards, and statistical error analysis.

C. Subtopic Map

1. Experimental Problem Formulation and Hypothesis
2. Variable Isolation and Experimental Setup
3. Data Collection, Measurement, and Tabulation
4. Graphical Representation and Curve Fitting
5. Statistical Analysis and Experimental Evaluation

D. Concept Graph

- Experimental Problem Formulation and Hypothesis: Testable problem statements identifying independent and dependent variables, directional hypothesis formulations (If \dots, then \dots, because \dots).
- Variable Isolation and Experimental Setup: Independent variable operational definition and levels, dependent variable operational definition, controlled constants identification, control group rationale.
- Data Collection, Measurement, and Tabulation: Qualitative vs quantitative observations, data table formatting with labeled units, measurement precision and significant figures.
- Graphical Representation and Curve Fitting: Graph axis selection and scaling, plot titling, line of best fit determination, equation of line ($y = mx + b$), coefficient of determination (R^2).
- Statistical Analysis and Experimental Evaluation: Mean, median, mode, range, standard deviation calculations, systematic vs random error analysis, realistic experimental improvement suggestions.

Forensics

B. Official 2027 scope

- Exact 2027 scope summary: Requires qualitative analysis of unknown chemical reagents, organic polymers, fibers, and physical crime scene evidence to solve a crime scenario.
- Official topics explicitly named: Qualitative inorganic chemical analysis, organic polymers and plastics, fiber analysis, chromatography, physical evidence, blood typing, and DNA profiling.

C. Subtopic Map

1. Qualitative Chemical Analysis of Unknown Solids and Liquids
2. Polymers, Plastics, and Fiber Identification
3. Chromatography and Chemical Spectroscopy
4. Serology, DNA Analysis, and Biological Evidence
5. Physical Evidence and Crime Scene Reconstruction

D. Concept Graph

- Qualitative Chemical Analysis of Unknown Solids and Liquids: Identification of white powders (NaHCO_3 , Na_2CO_3 , NaCl , CaSO_4 , cornstarch, glucose, sucrose), solubility rules, HCl effervescence testing, iodine starch testing, Benedict's reducing sugar testing, flame tests for metal cations (Na^+ , K^+ , Ca^{2+} , Cu^{2+} , Li^+).
- Polymers, Plastics, and Fiber Identification: Synthetic resin code identification (PETE, HDPE, PVC, LDPE, PP, PS) via burn behavior and density gradient testing, natural fibers (wool, cotton, silk) vs synthetic fibers (nylon, polyester, acrylic) microscopic burn behavior.
- Chromatography and Chemical Spectroscopy: Thin-layer chromatography (TLC) and paper chromatography of inks/dyes, retention factor (R_f) calculations, mass spectrometry and infrared (FTIR) spectrum interpretation basics.
- Serology, DNA Analysis, and Biological Evidence: ABO/Rh blood typing presumptive testing, Kastle-Meyer test, luminol chemiluminescence, gel electrophoresis DNA pattern analysis, Short Tandem Repeat (STR) profiling.
- Physical Evidence and Crime Scene Reconstruction: Fingerprint pattern classification (loops, whorls, arches) and minutiae matching, hair microscopic anatomy and medullary index calculation, glass fracture analysis (radial vs concentric, Becke line refractive index), shoeprint and tire tread casting.

Hovercraft

B. Official 2027 scope

- Exact 2027 scope summary: Integrates a theoretical physics written exam on fluid mechanics and mechanics principles with a hands-on hovercraft device flight performance trial.
- Official topics explicitly named: Fluid mechanics, air cushion pressure dynamics, impulse and momentum, friction, electric circuits, motor power.

C. Subtopic Map

1. Fluid Statics and Air Cushion Mechanics
2. Aerodynamic Forces and Fluid Dynamics
3. Classical Mechanics, Impulse, and Momentum
4. Device Electromechanical Systems
5. Performance Testing and Kinematic Calculations

D. Concept Graph

- Fluid Statics and Air Cushion Mechanics: Pascal's law, air cushion pressure differential ($\Delta P = F/A$), hovercraft lift equilibrium, cushion perimeter escape dynamics, ideal gas law air density calculations.
- Aerodynamic Forces and Fluid Dynamics: Bernoulli's principle ($P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$), Venturi effect, aerodynamic drag equation ($F_d = \frac{1}{2}\rho v^2 C_d A$), boundary layer friction, laminar vs turbulent air flow around lift/thrust fans.
- Classical Mechanics, Impulse, and Momentum: Newton's laws of motion applied to thrust, momentum conservation ($p = mv$), impulse-momentum theorem ($J = \int F dt = \Delta p$), kinetic energy vs electrical power ($P = F v$).
- Device Electromechanical Systems: DC motor circuit analysis, battery voltage degradation under load, rheostat and PWM speed control circuits, fan blade pitch aerodynamic efficiency.
- Performance Testing and Kinematic Calculations: Mass-to-lift performance ratios, target distance stopping dynamics, deceleration vector analysis.

Protein Modeling

B. Official 2027 scope

- Exact 2027 scope summary: Focuses on structural biology, amino acid chemistry, physical 3D model construction, and computer visualization, rotating specifically to CRISPR-Cas9 genome editing complexes for 2027.
- Official topics explicitly named: Amino acid chemical properties, levels of protein structure, CRISPR-Cas9 biochemistry, base editing modifications, PyMOL/Jmol visualization software.

C. Subtopic Map

1. Amino Acid Chemistry and Primary Structure
2. Protein Secondary, Tertiary, and Quaternary Folding
3. Molecular Visualization Software Commands (PyMOL / Jmol)
4. CRISPR-Cas9 Structure and Target Recognition Mechanics
5. Base Editing Mechanics and Cas9 Variant Engineering

D. Concept Graph

- Amino Acid Chemistry and Primary Structure: 20 standard amino acid chemical structures, side chain properties (hydrophobic, polar uncharged, positively charged,

negatively charged), peptide bond formation (N-terminus to C-terminus), isoelectric point (pI).

- Protein Secondary, Tertiary, and Quaternary Folding: Alpha helices and beta pleated sheets hydrogen bonding patterns, tertiary side chain interactions (disulfide bonds, salt bridges, hydrophobic collapse, van der Waals forces), Ramachandran plot angles (ϕ , ψ), quaternary multimeric assembly.
- Molecular Visualization Software Commands (PyMOL / Jmol): Command-line rendering scripts, PDB file header parsing, structural color schemes, distance measurements, hydrogen bond toggles.
- CRISPR-Cas9 Structure and Target Recognition Mechanics: Cas9 domain architecture (REC lobe, NUC lobe, RuvC, HNH endonuclease domains), Protospacer Adjacent Motif (PAM sequence 5'-NGG-3'), single guide RNA (sgRNA) spacer/scaffold mechanics, R-loop formation, double-strand break cleavage.
- Base Editing Mechanics and Cas9 Variant Engineering: Catalytically inactive dCas9 and nCas9 nickase engineering, Cytidine Base Editors (CBE: C to T conversions), Adenine Base Editors (ABE: A to G conversions), prime editing reverse transcriptase complexes.

Remote Sensing

B. Official 2027 scope

- Exact 2027 scope summary: Examines satellite remote sensing physics, electromagnetic radiation laws, sensor architectures, digital image processing, and climate monitoring applications.
- Official topics explicitly named: Electromagnetic spectrum, blackbody radiation laws, sensor orbits and resolution, digital image processing, vegetation spectral indices, climate applications.

C. Subtopic Map

1. Electromagnetic Radiation Physics
2. Satellite Orbits and Sensor Architectures
3. Digital Image Processing and Spatial Analysis
4. Spectral Signatures and Band Ratio Indices
5. Earth System Observation and Climate Applications

D. Concept Graph

- Electromagnetic Radiation Physics: Electromagnetic spectrum regions (UV, Visible, NIR, SWIR, TIR, Microwave), Planck's law, Wien's displacement law ($\lambda_{\max} = 2898 / T$), Stefan-Boltzmann law, atmospheric windows, atmospheric scattering (Rayleigh, Mie, Non-selective).
- Satellite Orbits and Sensor Architectures: Geostationary vs sun-synchronous polar orbits, active sensors (LiDAR, Radar) vs passive sensors (optical radiometers), spatial, spectral, radiometric, and temporal resolution trade-offs.
- Digital Image Processing and Spatial Analysis: Spatial filtering convolution kernels,

histogram contrast stretching, supervised (Maximum Likelihood) vs unsupervised (K-means) classification, Principal Component Analysis (PCA).

- Spectral Signatures and Band Ratio Indices: Vegetation, water, and soil spectral reflectance curves, Normalized Difference Vegetation Index ($\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$), Normalized Difference Water Index (NDWI), Normalized Burn Ratio (NBR).
- Earth System Observation and Climate Applications: Thermal infrared land surface temperature extraction, radar altimetry ice sheet mass balance, global forest carbon stock monitoring, flood and wildfire disaster mapping.

Rocks and Minerals

B. Official 2027 scope

- Exact 2027 scope summary: Evaluates mineralogy, crystallography, petrology (igneous, sedimentary, metamorphic), economic geology, and stratigraphic principles.
- Official topics explicitly named: Mineral identification, crystal systems, Bowen's Reaction Series, rock texture classification, ore deposits, and stratigraphic mapping.

C. Subtopic Map

1. Crystallography and Mineral Physics
2. Igneous Petrology and Magmatic Processes
3. Sedimentary Petrology and Stratigraphy
4. Metamorphic Petrology and Structural Geology
5. Economic Geology and Mineral Resources

D. Concept Graph

- Crystallography and Mineral Physics: 6 crystal systems (Isometric, Tetragonal, Hexagonal, Orthorhombic, Monoclinic, Triclinic), Mohs hardness scale, cleavage vs fracture patterns, specific gravity determination, optical properties (double refraction, fluorescence).
- Igneous Petrology and Magmatic Processes: Bowen's Reaction Series (discontinuous vs continuous branches), magma chemical compositions (felsic, intermediate, mafic, ultramafic), rock textures (aphanitic, phaneritic, porphyritic, pegmatitic, vesicular, glassy), intrusive vs extrusive landforms.
- Sedimentary Petrology and Stratigraphy: Clastic vs chemical/biochemical sedimentary rocks, grain size sorting and roundness (Wentworth scale), sedimentary structures (cross-bedding, ripple marks, graded bedding), stratigraphic principles (superposition, original horizontality, cross-cutting relationships).
- Metamorphic Petrology and Structural Geology: Metamorphic grade index minerals (chlorite, biotite, garnet, staurolite, kyanite, sillimanite), foliated (slate, phyllite, schist, gneiss) vs non-foliated textures, contact vs regional metamorphism, P-T phase diagrams.
- Economic Geology and Mineral Resources: Ore minerals (galena, chalcopyrite, hematite, sphalerite), hydrothermal vein deposits, placer deposits, industrial minerals, fossil fuel source rocks and traps.

Thermodynamics

B. Official 2027 scope

- Exact 2027 scope summary: Combines a written theoretical exam on thermal physics, heat transfer, and thermodynamic laws with an insulation device build evaluation.
- Official topics explicitly named: Laws of thermodynamics, heat capacity, heat transfer mechanisms (conduction, convection, radiation), thermodynamic cycles, calorimetry.

C. Subtopic Map

1. Temperature, Heat, and Thermal Expansion
2. Heat Transfer Mechanisms and Thermal Insulation
3. Laws of Thermodynamics and State Functions
4. Thermodynamic Processes and Ideal Cycles
5. Calorimetry and Thermal Measurement Techniques

D. Concept Graph

- Temperature, Heat, and Thermal Expansion: Temperature scale conversions (Kelvin, Celsius, Fahrenheit), linear and volumetric thermal expansion ($\Delta L = \alpha L_0 \Delta T$), specific heat capacity ($Q = m c \Delta T$), latent heat of phase changes ($Q = m L$).
- Heat Transfer Mechanisms and Thermal Insulation: Fourier's law of thermal conduction ($Q/t = k A \frac{\Delta T}{x}$), convective heat transfer, Stefan-Boltzmann law for radiation ($P = \epsilon \sigma A T^4$), insulation thermal resistance (R-value) multi-layer calculations.
- Laws of Thermodynamics and State Functions: Zeroth law, First law ($\Delta U = Q - W$), work done by expanded gases ($W = \int P dV$), enthalpy (H), entropy ($\Delta S = \int \frac{dQ}{T}$), Second and Third laws.
- Thermodynamic Processes and Ideal Cycles: Isothermal ($\Delta T = 0$), isobaric ($\Delta P = 0$), isochoric ($\Delta V = 0$), and adiabatic ($Q = 0$) processes, PV and TS diagrams, Carnot cycle maximum efficiency ($\eta = 1 - T_C / T_H$).
- Calorimetry and Thermal Measurement Techniques: Coffee-cup and bomb calorimetry equations, calorimeter heat capacity (C_{cal}), thermocouples, thermistors, RTD temperature sensors.

Water Quality

B. Official 2027 scope

- Exact 2027 scope summary: Evaluates aquatic ecology, water chemistry testing, and biological indicators, rotating specifically to Marine and Estuarine ecosystems for the 2027 season.
- Official topics explicitly named: Marine ecosystems, estuaries, water chemistry parameters, marine bioindicator species, estuarine hydrology, and salinometer physics.

C. Subtopic Map

1. Marine and Estuarine Ecosystem Dynamics
2. Water Chemistry Parameters and Quality Indices
3. Marine Bioindicators and Trophic Cascades
4. Estuarine Hydrology and Ocean Pollution
5. Salinometer/Hydrometer Physics and Calibration

D. Concept Graph

- Marine and Estuarine Ecosystem Dynamics: Estuary classifications (salt-wedge, partially mixed, well-mixed), estuarine zonation, salt marshes and mangrove ecology, intertidal stress adaptation, coral reef biomes, ocean food webs.
- Water Chemistry Parameters and Quality Indices: Dissolved oxygen (DO) saturation, Biochemical Oxygen Demand (BOD), seawater pH buffering capacity, nitrate, phosphate, and ammonia nutrient cycling, ocean salinity and chlorinity calculations ($\text{Salinity} = 1.80655 \times \text{Chlorinity}$), eutrophication.
- Marine Bioindicators and Trophic Cascades: Estuarine and marine macroinvertebrates, holoplankton vs meroplankton, indicator organisms for polluted vs clean marine waters, coral bleaching indicators.
- Estuarine Hydrology and Ocean Pollution: Point vs non-point source pollution, thermal pollution dynamics, estuarine flushing times, ocean microplastic accumulation, heavy metal biomagnification.
- Salinometer/Hydrometer Physics and Calibration: Archimedes' principle ($F_b = \rho_{\text{fluid}} V_{\text{submerged}} g$), density testing across 1-10‰ mass/volume salinity range, device construction geometry.

PART 2 — DATA MODEL RECOMMENDATION

To support automated question generation, drill execution, binder organization, missing topic audits, and weakness diagnostic algorithms on SciOly.app, the system data model must implement a relational database architecture.

Database Schema Structure

The database schema links multi-season event configurations to official topic buckets, study subtopics, granular concept nodes, and individual student mastery tracking.

The relational hierarchy follows a strict parent-to-child flow:

- seasons (1 to N) $\rightarrow$ events
- events (1 to N) $\rightarrow$ topics
- topics (1 to N) $\rightarrow$ subtopics
- subtopics (1 to N) $\rightarrow$ concepts
- concepts (1 to N) $\rightarrow$ question_concepts

- concepts (1 to N) \to student_concept_mastery

-- Season Table

```
CREATE TABLE seasons (  
    season_id INT PRIMARY KEY,  
    year INT NOT NULL UNIQUE  
);
```

-- Event Table

```
CREATE TABLE events (  
    event_id VARCHAR(64) PRIMARY KEY,  
    season_id INT REFERENCES seasons(season_id),  
    division CHAR(1) CHECK (division IN ('B', 'C')),  
    name VARCHAR(128) NOT NULL,  
    domain VARCHAR(64) NOT NULL,  
    test_type VARCHAR(64) NOT NULL,  
    rotation_topic VARCHAR(128),  
    allowed_materials_summary TEXT  
);
```

-- Topic Table (Official Rules Buckets)

```
CREATE TABLE topics (  
    topic_id VARCHAR(64) PRIMARY KEY,  
    event_id VARCHAR(64) REFERENCES events(event_id),  
    title VARCHAR(256) NOT NULL,  
    official_rule_reference VARCHAR(64)  
);
```

-- Subtopic Table (Practicable Study Units)

```
CREATE TABLE subtopics (  
    subtopic_id VARCHAR(64) PRIMARY KEY,  
    topic_id VARCHAR(64) REFERENCES topics(topic_id),  
    title VARCHAR(256) NOT NULL,  
    display_order INT  
);
```

-- Concept Table (Atomic Question Targets)

```
CREATE TABLE concepts (  
    concept_id VARCHAR(64) PRIMARY KEY,  
    subtopic_id VARCHAR(64) REFERENCES subtopics(subtopic_id),  
    title VARCHAR(256) NOT NULL,  
    description TEXT,  
    depth_level INT DEFAULT 1  
);
```

-- Question Tagging Junction Table

```
CREATE TABLE question_concepts (  
    question_id VARCHAR(64) NOT NULL,  
    concept_id VARCHAR(64) REFERENCES concepts(concept_id),
```

```

        PRIMARY KEY (question_id, concept_id)
    );

-- Student Mastery Tracking Table
CREATE TABLE student_concept_mastery (
    user_id VARCHAR(64) NOT NULL,
    concept_id VARCHAR(64) REFERENCES concepts(concept_id),
    attempts INT DEFAULT 0,
    correct INT DEFAULT 0,
    mastery_score FLOAT DEFAULT 0.0,
    last_practiced TIMESTAMP,
    PRIMARY KEY (user_id, concept_id)
);

```

System Granularity Recommendations

1. Ideal Granularity for Question Tagging

Every practice question in the SciOly.app item bank should be tagged at the **Concept level** with exactly one primary concept ID and up to two optional secondary concept IDs. Tagging at the broader Subtopic level fails to diagnose specific knowledge gaps (for example, distinguishing between a student's failure in cardiac conduction math versus heart valve anatomy). Tagging at a hyper-microscopic level (such as individual star names or one-off facts) creates database fragmentation. Anchoring questions to repeatable concept nodes balances diagnostic utility with item bank maintenance.

2. Ideal Granularity for Binder Chapters

Binder generation and audit tools should operate at the **Subtopic level**. A competitive Science Olympiad study binder should contain 4 to 6 tabbed subtopic sections per event. Organizing binders at the subtopic level provides clear structural divisions that align with standard 50-minute exam time constraints, allowing competitors to locate reference material rapidly during competition.

3. Ideal Granularity for Intervention and Weakness Repair

Weakness diagnostics and automated drill generation should evaluate performance by aggregating student performance across all concepts within a **Subtopic**. When a student's accuracy within a subtopic drops below 65%, SciOly.app should flag the entire subtopic as a weakness area and automatically queue targeted drill sets drawing from the underlying low-scoring concepts.

4. Event Taxonomy Architecture Assignment

Different Science Olympiad events require varying database taxonomy tree structures depending on their theoretical depth, rotational variability, and computational nature.

Taxonomy Structure Type	Applicable Events	Structural Rationale
Deep Concept Trees (3-Tier: Topic ↳to Subtopic ↳to Concept)	Anatomy and Physiology, Astronomy, Botany, Chemistry Lab, Circuit Lab, Designer Genes, Disease Detectives, Dynamic Planet, Forensics, Protein Modeling, Remote Sensing, Rocks and Minerals, Thermodynamics, Water Quality	Events driven by extensive memorization, biochemical pathways, physical formulas, and rotational topic shifts require deep 3-tier concept mapping for diagnostic weakness tagging.
Shallow Topic Trees (2-Tier: Topic ↳to Subtopic)	Engineering CAD, Codebusters, Experimental Design, Hovercraft	Events governed by standardized scoring rubrics, algorithmic cipher execution, or hardware performance tests require shallow topic structures to prevent over-engineering.

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